Telco-Customer Churn Analysis Report — ABC Communications Ltd

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Business Understanding Report — Customer Churn Analysis for ABC Communications Ltd

The Business Problem

Customer churn — when a subscriber cancels or switches away from a company's service — is one of the most expensive problems a subscription-based business like a telecom provider can face. Industry research consistently shows that acquiring a new customer cost significantly more than retaining an existing one, often estimated at 5–7 times more. For ABC Communications Ltd, every customer who churns doesn't just represent lost monthly revenue; it represents a lost investment in acquisition, onboarding, and service delivery that will never be recovered.

In competitive telecom markets, churn is rarely random. It tends to cluster around specific, identifiable patterns — certain contract types, certain service tiers, certain stages of the customer relationship. This makes churn a solvable problem, not just an unavoidable cost of doing business: if a company can identify which customers are most likely to leave and why, it can intervene before they do.

Why This Analysis Matters

This project investigates 7,043 ABC Communications customer records to understand the shape and drivers of churn across the business. Rather than treating all customers as equally at risk, the analysis breaks the customer base down by contract type, tenure, service usage, payment method, and demographics to identify where churn risk actually concentrates.

The findings are intended to move the business from a reactive posture (losing customers and not knowing why) to a proactive one (identifying at-risk segments and acting before cancellation).

Scope of the Analysis

The dataset covers customer-level records including demographic details (gender, senior citizen status, partner/dependent status), account information (tenure, contract type, payment method, monthly and total charges), and subscribed services (phone, internet, and add-ons like online security, tech support, and streaming). The analysis examines how each of these factors relates to the customer's churn status, using pivot analysis, visualizations, and correlation analysis to surface the strongest relationships in the data.

Expected Business Value

By the end of this analysis, ABC Communications will have a clear, data-backed picture of: which customer segments carry the highest churn risk, which business practices (contract structuring, service bundling, payment method encouragement) most influence retention, and where the company's retention efforts and incentives would have the greatest impact. This directly supports the goal of reducing churn and protecting recurring revenue.

Data Inspection Report

1. Dataset Overview

• Source: Telco Customer Churn Dataset (Kaggle)

• Total records: 7,043

• Total columns: 21 original (26 after your calculated columns)

• Granularity: one row per customer

2. Column Inventory

Briefly list the 21 original columns grouped by type:

• Identifiers: customerID

• Demographics: gender, SeniorCitizen, Partner, Dependents

• Account info: tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges

• Services: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

• Target variable: Churn

3. Data Quality Findings

• Missing values: 11 rows (0.16%) had blank TotalCharges values

• Root cause: all 11 rows had tenure = 0 — brand-new customers not yet billed.

• Duplicates: 0 duplicate rows found

• Data type issues: TotalCharges was stored as text due to the blank entries; SeniorCitizen was stored as 0/1 instead of Yes/No.

4. Cleaning Actions Taken

• Converted TotalCharges to numeric (Decimal Number) and filled the 11 blanks with 0, since tenure=0 means no charges have accrued yet — a logical zero, not an unknown value

• Converted SeniorCitizen from 0/1 to Yes/No for consistency with other binary fields.

• Verified all remaining column types matched their expected format (text categories, whole numbers, decimals).

5. Summary Statistics

• Average tenure: 32.4 months

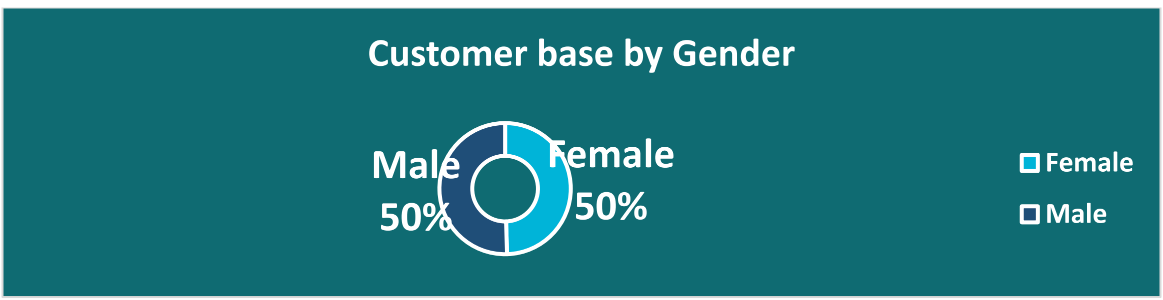
• Average MonthlyCharges: $64.76

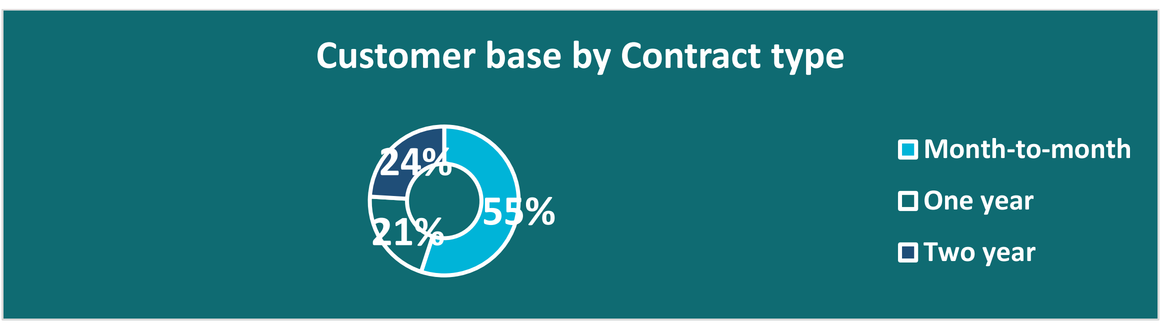
• Churn rate: 26.5%

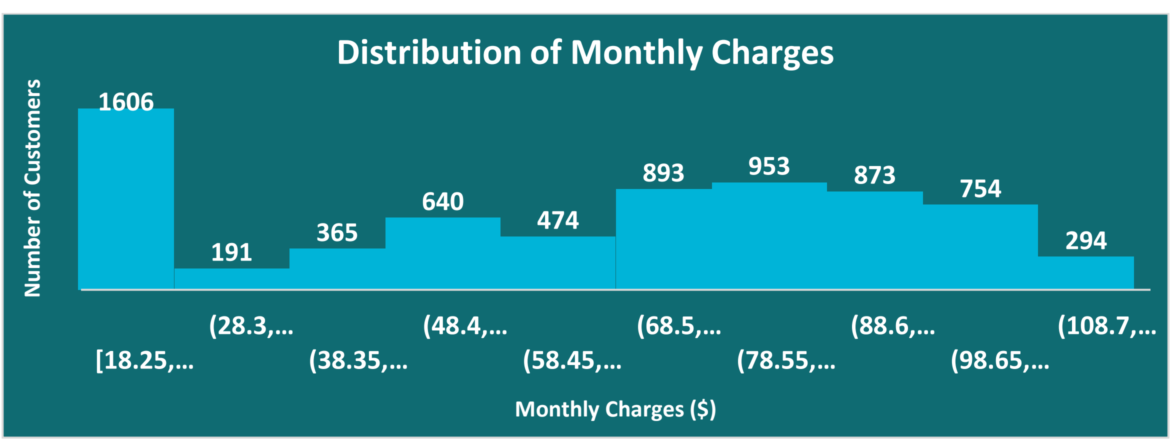
Business Data Analysis

Q1: What does the customer base look like?

ABC Communications' customer base consists of 7,043 subscribers. The base is evenly split by gender, with 50% female and 50% male customers, indicating gender is not a meaningful differentiator in this analysis. Contract distribution, however, reveals an important pattern: 55% of customers are on month-to-month contracts, while only 21% hold one-year contracts and 24% hold two-year contracts. This means the majority of the customer base sits on the least stable contract type, a point that becomes significant in later sections. The average customer has been with the company for 32.4 months and pays $64.76 per month. The distribution of monthly charges shows that a large cluster of customers pay in the lower range (around $20), consistent with basic single-service plans, while the remaining base is spread more broadly through the $60-$100 range, reflecting bundled multi-service subscriptions.





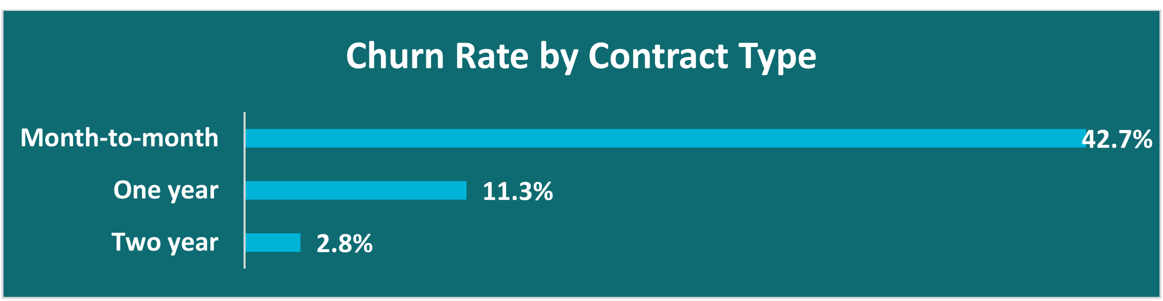


Q2: Which segments have the highest churn?

Several customer segments show meaningfully elevated churn rates. Senior citizens churn at 41.7%, nearly double the rate of non-senior customers at 23.6%. Customers without a partner churn at 33.0%, compared to 19.7% for those with a partner. Similarly, customers without dependents churn at 31.3%, versus 15.5% for those with dependents. By contrast, gender shows almost no relationship with churn (26.9% for female customers versus 26.2% for male customers), indicating it is not a useful segmentation factor for retention strategy. Overall, the data suggests that customers with fewer household or family ties — no partner, no dependents, and particularly senior citizens — represent the highest-risk segments for churn.

Q3: Does contract type influence retention?

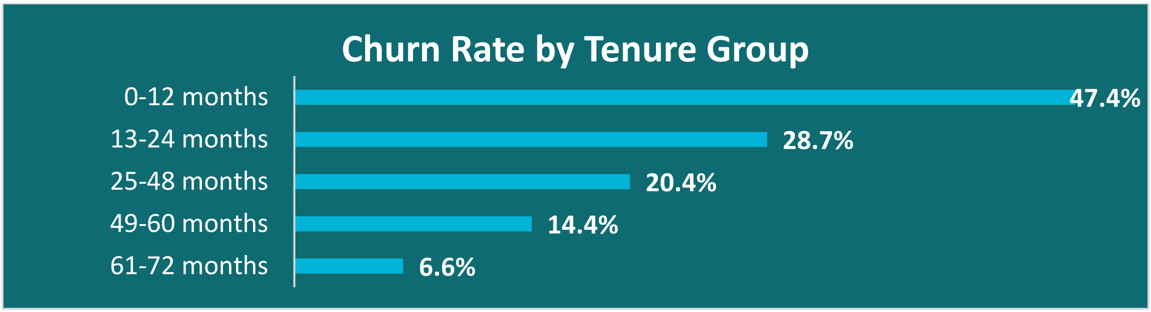
Contract type is the single strongest predictor of churn identified in this analysis. Month-to-month customers churn at a rate of 42.7%, compared to 11.3% for one-year contracts and just 2.8% for two-year contracts — a nearly fifteen-fold difference between the two extremes. This finding is particularly significant given that 55% of ABC Communications' customer base currently holds a month-to-month contract, as shown in the Q1 analysis. This means the majority of the company's customers sit in its highest-risk contract category, representing a substantial exposure to preventable revenue loss. Encouraging migration toward longer-term contracts represents one of the clearest, most actionable opportunities available to reduce overall churn.

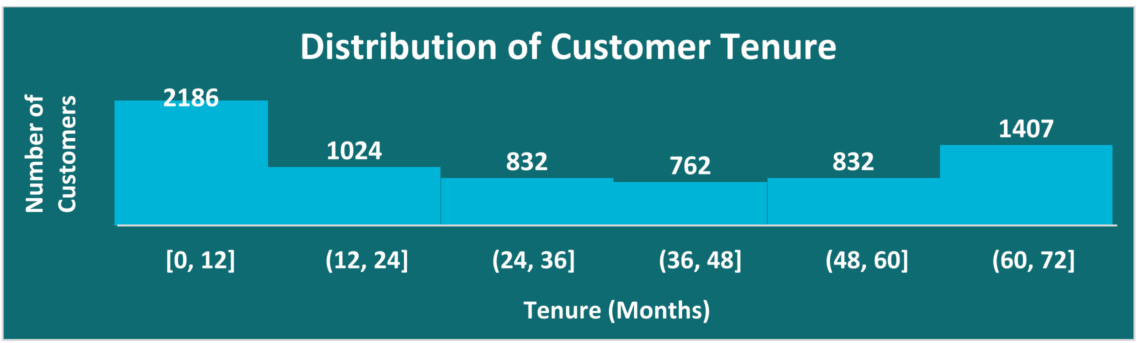


Q4: Does tenure affect loyalty?

Customer tenure shows a clear and consistent relationship with loyalty. Churn is highest among customers in their first year (47.4%), then steadily declines across each subsequent tenure band: 28.7% for 13-24 months, 20.4% for 25-48 months, 14.4% for 49-60 months, and just 6.6% for customers who have stayed 61-72 months. This steady decline indicates that the longer a customer remains with the company, the less likely they are to leave — loyalty compounds over time rather than staying constant.

The distribution of tenure across the customer base adds further context: rather than a smooth spread, the data shows two distinct clusters — a large group of relatively new customers (0-12 months) and a second, smaller cluster of long-term customers (61-72 months), with fewer customers in between. This suggests ABC Communications effectively has two different customer populations to manage: a large group of newer, higher-risk customers who need active retention effort, and a stable base of loyal long-term subscribers.





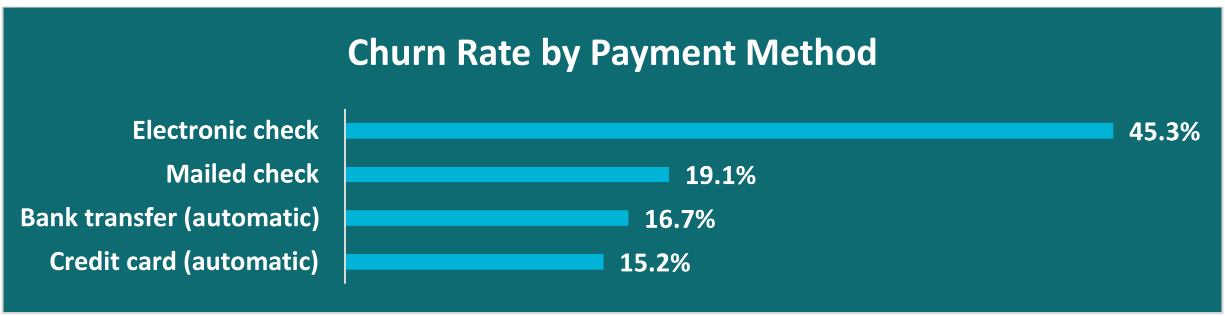
Q5: Which services influence churn?

Service type and usage show meaningful relationships with churn. Customers with Fiber optic internet churn at 41.9%, more than double the rate of DSL customers (19.0%) and far above customers with no internet service at all (7.4%) — a notable finding given that fiber is typically the company's premium offering. Lack of protective and support add-on services also correlates strongly with churn: customers without Online Security churn at 41.8% compared to 14.6% for those who have it, and customers without Tech Support churn at 41.6% compared to 15.2% for those who have it.

Interestingly, the relationship between the number of add-on services a customer holds and their likelihood of churning is not straightforward. Churn actually peaks among customers with exactly one add-on service (45.8%) — higher than customers with no add-ons at all (21.4%) — before declining steadily as service count increases, reaching just 5.3% among customers with all six services. This suggests that partial service adoption may be riskier than either no adoption or full adoption, possibly because customers who have "tried" a single service without further engagement are less invested in the platform overall.

Q6: Which payment methods have higher churn?

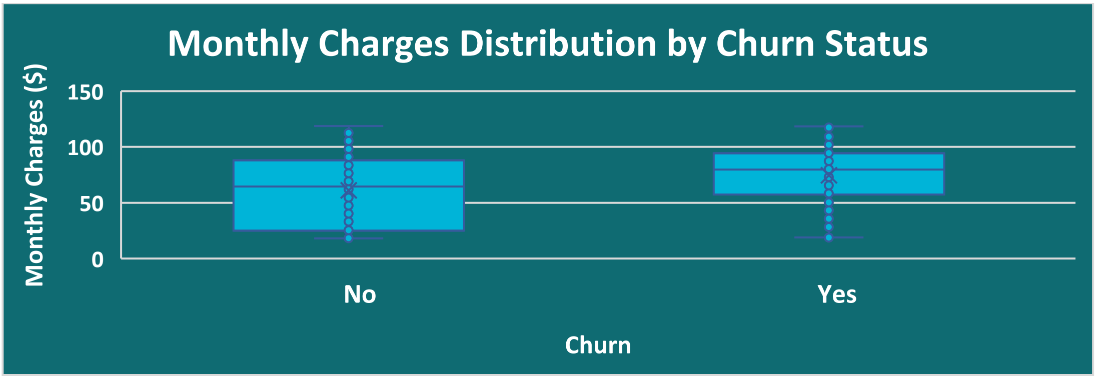
Payment method shows one of the largest churn disparities in the dataset. Customers paying by electronic check churn at 45.3%, substantially higher than mailed check (19.1%), automatic bank transfer (16.7%), or automatic credit card payment (15.2%). This is particularly significant because electronic check is also the most common payment method in the customer base, used by 2,365 customers — meaning this high-churn pattern affects a large share of the business rather than a small edge case. The consistently lower churn among all three automatic payment methods suggests that payment automation itself may be associated with stronger customer retention, whether through reduced friction, increased convenience, or simply attracting more committed subscribers.



Cross-Factor Analysis: Spend Patterns and Correlation

To understand these relationships further, monthly charges were compared directly against churn status, and a correlation matrix was built across the key numeric variables in the dataset. Customers who churned show a noticeably higher median monthly charge than customers who stayed, reinforcing the earlier finding that higher-cost customers — particularly fiber optic subscribers — are more likely to leave.

The correlation analysis confirms tenure as the strongest individual factor associated with churn (a correlation of -0.35), meaning churn risk decreases consistently the longer a customer stays. Monthly charges show a weaker positive relationship with churn (+0.19), while service count shows only a weak relationship (-0.09) when measured as a straight-line correlation — a result that does not contradict the earlier finding, since correlation only captures linear patterns and cannot detect the non-linear "peak at one service" pattern identified in the Q5 analysis. Notably, tenure and total charges are highly correlated with each other (0.83), indicating that total charges largely reflect how long a customer has stayed rather than how much they spend per month — an important distinction for interpreting the data correctly.



Correlation Heatmap

	Tenure	Monthly Charges	Total Charges	Service Count	Churn Flag
Tenure	1	0.247899856	0.826178398	0.494262611	-0.352229
Monthly Charges	0.2479	1	0.651173832	0.72470601	0.193356
Total Charges	0.82618	0.651173832	1	0.744827084	-0.198324
Service Count	0.49426	0.72470601	0.744827084	1	-0.087698
Churn Flag	-0.3522	0.193356422	-0.198324263	-0.08769813	1

Q7: What actions should management take?

The patterns identified across contract type, tenure, service usage, payment method, and customer demographics point toward a set of specific, actionable strategies for reducing churn. These are detailed in full in the Business Recommendations section of this report.

Insights, Risks and Opportunities

• Insights (data-backed observations)

1. Contract type is the single strongest churn driver — Month-to-month customer's churn at 42.7%, versus 11.3% for One-year and just 2.8% for Two-year contracts. A ~15x gap between the extremes.

2. The first year is the highest-risk window — churn drops steadily and consistently from 47.4% (0-12 months) down to 6.6% (61-72 months). Customers who survive their first year become dramatically more loyal.

3. Electronic check users churn far more than other payment methods — 45.3%, nearly triple the rate of automatic bank transfer or credit card users (~15-17%). It's also your largest payment segment (2,365 customers), making it high-impact.

4. Premium service doesn't guarantee loyalty — Fiber optic internet customers churn at 41.9%, more than double DSL customers (19.0%), despite being the higher-end product.

5. Add-on services correlate with retention, but non-linearly — churn actually peaks among customers with just 1 add-on service (45.8%), higher than customers with zero add-ons (21.4%), then steadily drops as service count rises toward 6 (5.3%). Partial adoption looks riskier than no adoption or full adoption.

• Business Risks

1. Revenue concentration in a high-risk segment — 55% of the customer base is on month-to-month contracts, the exact group with the highest churn rate. This is a large share of revenue sitting on the least stable footing.

2. Fiber optic churn threatens the premium product line — losing customers on your highest-value service tier is more costly per lost customer than losing a basic-plan customer, since fiber customers likely carry higher MonthlyCharges.

3. Payment friction may be silently driving cancellations — the electronic check pattern suggests either a billing experience problem or that this payment method attracts a less committed customer segment in the first place; either way, it's an unaddressed risk sitting in your largest payment group.

- Opportunities

1. Contract conversion campaigns — incentivizing month-to-month customers to switch to annual contracts (discount, bundled perks) could directly cut churn given the enormous rate gap already documented.

2. First-year retention program — since churn risk is highest and steadily declining in year one, a structured onboarding/check-in program during the first 12 months could catch at-risk customers before they leave.

3. Payment method migration — nudging electronic check users toward autopay (bank transfer/credit card) could reduce churn in your largest payment segment, potentially through a small autopay discount incentive.

Business Recommendation

1. Incentivize contract upgrades for month-to-month customers

Offer a modest discount (e.g., 10-15% off monthly rate) or a free add-on service (like OnlineSecurity or TechSupport) for customers who switch from month-to-month to a One- or Two-year contract. Given the 42.7% → 2.8% churn gap, even converting a fraction of this 3,875-customer segment would meaningfully cut overall churn.

2. Launch a structured first-year retention program

Since 47.4% of churn happens within the first 12 months, implement proactive check-ins at 30, 90, and 180 days — usage reviews, satisfaction surveys, or personalized service recommendations. This targets churn at its steepest point rather than reacting after the fact.

3. Investigate and address the electronic check churn pattern

Before assuming it's purely a payment-method issue, investigate whether it reflects a billing experience problem (manual payment friction, late fees, failed payments) or a customer profile issue (this payment method may simply attract less price-committed customers). Pair the investigation with an incentive to switch to autopay (bank transfer/credit card), which show 15-17% churn versus 45.3%.

4. Review the fiber optic customer experience

Since fiber optic churn (41.9%) is more than double DSL (19.0%) despite being the premium product, investigate whether this stems from pricing perception, service reliability/outages, competitive pressure in fiber markets, or unmet expectations for a "premium" tier. A targeted customer satisfaction survey for fiber subscribers specifically would clarify the root cause before designing a fix.

5. Bundle add-on services more deliberately, especially for single-service adopters

Since churn peaks among customers with exactly 1 add-on service (45.8%) — higher than customers with none — consider restructuring how add-ons are sold. Rather than upselling services individually, bundle 2-3 complementary services (e.g., OnlineSecurity +TechSupport) as a package, which may explain why customers with 5-6 services show the lowest churn (5.3-12.4%) — full adoption creates stickiness that partial adoption doesn't.

6. Prioritize retention spend using CLTV

With total estimated customer lifetime value at $16,055,091 across the base, use the CLTV Estimate column to identify which at-risk customers (month-to-month, fiber, electronic check) represent the highest lifetime value, and prioritize retention outreach and incentive budget toward them first rather than spreading efforts evenly across all at-risk customers.